

	results in the satellite accelerating continuously towards the earth, as the direction of the velocity is constantly changing.
2(a)	$x = \frac{\lambda D}{a} \Rightarrow (1.4 \times 10^{-3}) = \frac{\lambda(2.1)}{(0.95 \times 10^{-3})}$ $\lambda = 6.3 \times 10^{-7} \text{ m}$
2(b)	Intensity decreases means that the amplitude of the wave decreases. Hence the intensity of the bright fringes decreases. Also, with the difference in amplitudes of wave from both slits, there will not be complete destructive interference. As a result, the contrast between the bright and dark fringes decrease. There is no change in the fringe separation.
2(c)	When the slits are widened, there waves are less diffracted as they pass through the slits. If the wavefronts do not overlap at the screen, there will be no interference, hence fringes are not observed.
3(a)	p.d. across $1.5 \text{ k}\Omega$ resistor = $(1.60 \times 10^{-3})(1.50 \times 10^3) = 2.40 \text{ V}$